Takuya Koriyama

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Research Interests

I'm broadly interested in statistics, ML, and probability theory. My previous research has focused on high-dimensional statistics, high-dimensional probability/geometry, and combinatorial stochastic processes. More recently, I'm working on applied AI research, including AI alignment and model evaluation.

Education

University of Chicago, Booth School of Business 2024–present
PhD student, Econometrics and Statistics. Advisor: Tengyuan Liang

Rutgers University 2022–2024
PhD student, Statistics. Advisor: Pierre C. Bellec

University of Tokyo 2018–2022
BE, Mathematical Engineering

Selected Publications

A Markov Chain Approach to Preference Alignment

Takuya Koriyama and Tengyuan Liang
Preprint, arXiv:2606.22652, 2026 [arXiv]

Honorable Mention for New Solution Concept, ICML 2026 NExT-Game Workshop

Introduces Markov Chain from Human Feedback to align generative models directly from pairwise preferences. Establishes convergence guarantees that adapt to non-transitive preferences and connects the method to reinforcement learning and game-theoretic alignment.

Precise Asymptotics of Bagging Regularized M-estimators

Takuya Koriyama, Pratik Patil, Jin-Hong Du, Kai Tan, and Pierre C. Bellec
The Annals of Statistics, 2026 [arXiv] [Journal]

Selected for the ICML Journal-to-Conference track

Derives precise prediction-error formulas and a consistent risk estimator for ensembles trained on overlapping subsamples. Shows how jointly tuning subsample size, ensemble size, and regularization can improve predictive performance.

Asymptotic Inference for Exchangeable Gibbs Partitions

Takuya Koriyama
Stochastic Processes and their Applications, 2026 [arXiv] [Journal]

Develops parameter estimation and predictive uncertainty quantification for random partition models, including asymptotically valid confidence intervals and error quantification for predicting allocation to existing or new groups.

Denoising Diffusions with Optimal Transport: Localization, Curvature, and Multi-Scale Complexity

Tengyuan Liang, Kulunu Dharmakeerthi, and **Takuya Koriyama**
Transactions on Machine Learning Research, 2026 [arXiv] [Journal]

Selected for Journal-to-Conference certification

Characterizes score-based denoising through optimal transport and quantifies how data geometry across noise scales determines denoising difficulty. Identifies bottleneck noise levels for complex, non-log-concave distributions.

Corrected generalized cross-validation for finite ensembles of penalized estimators

Pierre C. Bellec, Jin-Hong Du, **Takuya Koriyama**, Pratik Patil, and Kai Tan (alphabetical)
Journal of the Royal Statistical Society: Series B, 87(2), 289–318, 2025 [arXiv] [Journal]

Develops corrected generalized cross-validation for estimating prediction error in ensembles of penalized least-squares models. Provides statistical guarantees for an efficient evaluation and tuning method that avoids holding out training data or repeatedly refitting models.

Additional Publications

Asymptotics of resampling without replacement in robust and logistic regression
Pierre C. Bellec and **Takuya Koriyama** (alphabetical)
Bernoulli, 32(4), 3063–3087, 2026 [arXiv] [Journal]

Asymptotic mixed normality of maximum likelihood estimator for Ewens–Pitman partition
Takuya Koriyama, Takeru Matsuda, and Fumiyasu Komaki
Advances in Applied Probability, 58(1), 214–234, 2026 [arXiv] [Journal]

Phase transitions for the existence of unregularized M-estimators in single index models
Takuya Koriyama and Pierre C. Bellec
International Conference on Machine Learning (ICML), 2025 [arXiv] [Proceedings]

Error estimation and adaptive tuning for unregularized robust M-estimator
Pierre C. Bellec and **Takuya Koriyama** (alphabetical)
Journal of Machine Learning Research, 26(16), 1–40, 2025 [arXiv] [Journal]

Fully Data-driven Normalized and Exponentiated Kernel Density Estimator with Hyvärinen Score
Shunsuke Imai, **Takuya Koriyama**, Shouto Yonekura, Shonosuke Sugasawa, and Yoshihiko Nishiyama
Journal of Business & Economic Statistics, 43(1), 110–121, 2025 [arXiv] [Journal]

Existence of solutions to the nonlinear equations characterizing the precise error of M-estimators
Pierre C. Bellec and **Takuya Koriyama** (alphabetical)
Preprint, arXiv:2312.13254, 2023; submitted [arXiv]

Experience

Student Researcher, RIKEN Center for Brain Science 2021–2022
Theory of combinatorial stochastic processes and its application to network data analysis

Awards and Honors

- Honorable Mention for New Solution Concept, ICML NExT-Game Workshop, 2026.
- IMS Hannan Graduate Student Travel Award, 2025.
- Travel Award, Yale FDS Workshop Honoring Andrew Barron, 2024.
- Student Travel Award, IMS International Conference on Statistics and Data Science, 2022.
- Best Presentation Award, 16th Japan Statistical Society Spring Meeting, 2022.
- Japan Statistical Society Certificate Director's Award, 2022.

Academic Service

Reviewer: ICML; The Annals of Statistics; Journal of the American Statistical Association; Information and Inference: A Journal of the IMA.